



SGGS Institute of Engineering and Technology, Nanded

Summer Term Examination 2024-25

SY B Tech

Circuit Theory

Instrumentation Engineering

PCC-IN203

Max. Marks: 100

Date: 25-Jun-25

Time: 14.30 To 17.30 hrs

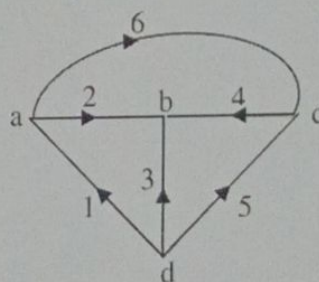
Notes:

1. All questions are compulsory.
2. Use of non programmable calculator is permitted
3. This paper contains two pages.
4. Assume suitable data, if necessary.
5. Figures to right indicate full marks.

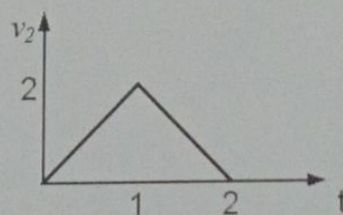
Marks

Q.1 Attempt any four:

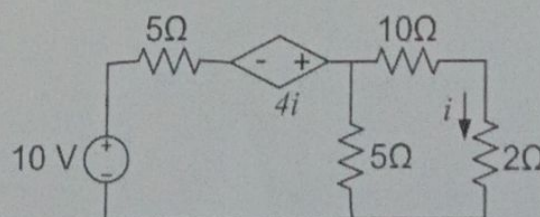
- a Define and explain: (5)
(i) Current, (ii) Duality, (iii) Planar graph, (iv) Unit ramp function and (v) Poles of a network function.
- b State and explain superposition theorem used for network analysis. (5)
- c For the graph shown, determine the tie-set matrix. (5)



- d State and explain final value theorem. (5)
- e For the waveform shown, write equation for $v_2(t)$ and find its transform $V_2(s)$. (5)

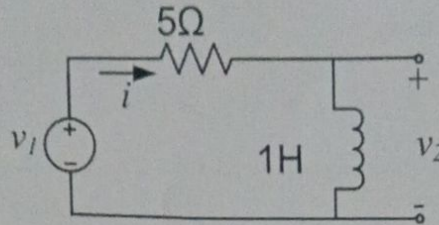


- Q.2 a What is the concept of complex frequency? Explain it with different special cases. (10)
- b For the network shown, find the values of current i . (10)



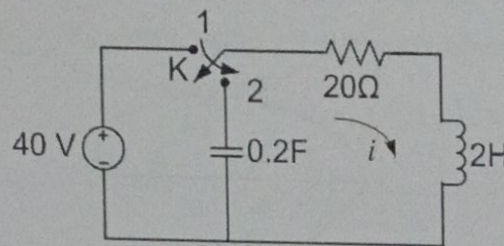
Q.3 a State and explain Millman's theorem with suitable example. (10)

b For the network shown, determine the transfer function $G_{12}(s) = \frac{V_2(s)}{V_1(s)}$. (10)

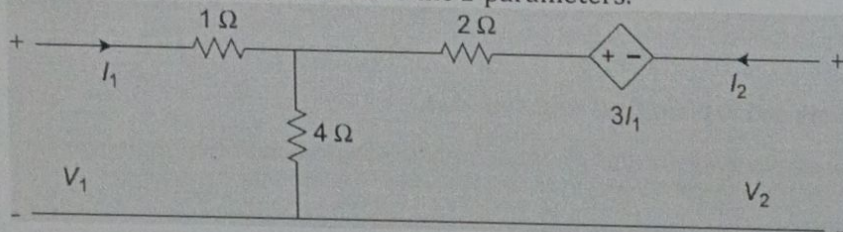


Q.4 a Explain the time domain behaviour from the locations of poles and zeros of network function. (10)

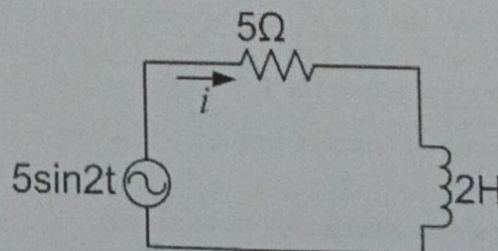
b In the network shown, switch K is changed from position '1' to position '2' at $t = 0$, a steady state having previously been established at position '1'. Solve for i , $\frac{di}{dt}$ and $\frac{d^2i}{dt^2}$ at $t = 0+$. (10)



Q.5 a For the network shown, determine the z-parameters. (10)



b For the network shown, determine the sinusoidal steady state current $i(t)$. (10)



//End of Paper//